

- Q.25 Explain the concept of time-weighted average (TWA) exposure limits in occupational exposure monitoring.
- Q.26 How can noise hazards be controlled in industrial workplaces?
- Q.27 Explain Concept of work station and its design.
- Q.28 Discuss the potential health effects of exposure to chemical solvents in industrial settings.
- Q.29 Benefits of biological monitoring.
- Q.30 What are the primary routes of exposure for occupational diseases in industrial settings?
- Q.31 Describe the symptoms and health effects of occupational asthma caused by exposure to workplace allergens and irritants.
- Q.32 Lung function test.
- Q.33 Describe the symptoms and long-term consequences of occupational dermatitis.
- Q.34 Initial medical examination and periodic medical examination.
- Q.35 Discuss the role of respiratory protective equipment occupational respiratory diseases.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Discuss the challenges associated with managing occupational exposure to hazardous chemicals and propose strategies for chemical hazards communication and control.
- Q.37 Explain the principles of ergonomic design and discuss how ergonomic interventions can prevent musculoskeletal disorders in the workplace.
- Q.38 Describe the methods used for assessing heat stress in the workplace, including wet bulb globe temperature (WBGT) measurements, heat stress indices, and physiological monitoring techniques.

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Roll No.

1st Year / Industrial Safety

Subject:- Industrial Hygiene and Occupational Health

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What term describes the physiological response to long-term exposure to high noise levels, resulting in a gradual loss of hearing sensitivity? Acoustic trauma
- Tinnitus
 - Noise-induced hearing loss
 - Hyperacusis
- Q.2 What is the primary objective of lockout/tagout procedures in industrial hygiene?
- Preventing unauthorized access to hazardous areas
 - Minimizing production downtime
 - Reducing energy consumption
 - Ensuring worker safety during equipment servicing
- Q.3 This is the preferred method of air sampling to evaluate worker exposure to airborne chemicals?
- personal
 - grab
 - area
 - integrated
- Q.4 Symptoms of this condition are fatigue, weakness, blurred vision, dizziness and headache?
- heat stress
 - heat exhaustion
 - dehydration
 - heat stroke
- Q.5 Which of the following is a common symptom of heat stress in the workplace?
- Hypothermia

- b) Elevated blood pressure
 - c) Excessive Sweating
 - d) Decreases heart rate
- Q.6 What term describes the process of identifying and assessing potential hazards associated with specific job tasks or processes to implement appropriate control measures?
- a) Hazard substitution b) Risk assessment
 - c) Job hazard analysis d) Exposure monitoring
- Q.7 What is the primary purpose of water purification in industrial hygiene?
- a) To reduce exposure to airborne contaminants
 - b) To ensure the safety of drinking water
 - c) To prevent exposure to chemical hazards
 - d) To control exposure to noise hazards
- Q.8 Which of the following occupational disease is also known as black lung disease?
- a) Byssinosis b) Silicosis
 - c) Pneumoconiosis d) Asbestosis
- Q.9 Which of the following is a common symptom of occupational asthma?
- a) Skin rash b) Eye irritation
 - c) Shortness of breath d) Headache
- Q.10 What are the 4 basic principles of industrial hygiene?
- a) Anticipation, Recognition, Evaluation, Control
 - b) Anticipation, recognition, Evaluation, Command
 - c) Anticipation, Recollection, Evaluation, Control
 - d) Anticipation, Recognition, Estimation, Control

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 _____ is the term for the physiological response to long-term exposure to high noise level, resulting in a gradual loss of hearing sensitivity?

- Q.12 Full form of ACGIH is _____.
- Q.13 _____ is the primary route of exposure for occupational diseases caused by inhalation of airborne contaminants?
- Q.14 _____ are synthetic or naturally occurring substances that damage, destroy, or impair the functioning of the central and/or peripheral nervous system.
- Q.15 Air sampling is a method used to measure the concentration of airborne contaminants in the workplace to assess exposure levels.
- a) True b) False
- Q.16 Nuisance can be defined as dust that contains less than 2% quartz
- a) True b) False
- Q.17 _____ means electromagnetic or corpuscular radiation capable of producing ions directly or indirectly in its passage through matter.
- Q.18 _____ is a condition characterized by inflammation of the lungs caused by inhalation of silica dust particles.
- Q.19 Allergens are substances that have the potential to cause allergic reactions in sensitive individuals upon exposure?
- a) True b) False
- Q.20 Neurotoxicants are substances that have the potential to cause harm to the nervous system, including neurological disorders, cognitive impairments, or behavioral changes?
- a) True b) False

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Discuss the role of personal protective equipment (PPE) in industrial hygiene
- Q.22 List three common physical hazards found in industrial environments
- Q.23 Describe the hierarchy of controls in industrial hygiene.
- Q.24 Describe the difference between threshold limit values (TLVs) and permissible exposure limits (PELs) in industrial hygiene.